

3. The photograph shows an insect called an aphid (*Aphis*).



Aphids damage crops such as barley, by making holes in the leaves. They then suck out sugar solution through the holes. A thick layer of fungi can then grow on the damaged leaves, so they absorb less light energy.

Farmers may use pesticides on their crops. Pesticides are effective, but may also be toxic to harmless organisms. Another approach is to release insects such as ladybirds onto the crop. Ladybirds are secondary consumers that are common in many food chains. They target pests such as aphids and so reduce their numbers.

(a) Using **only** the information above, give the evidence that:

(i) barley is photosynthetic; [1]

.....

(ii) aphids are primary consumers; [1]

.....

(iii) ladybirds are carnivores; [1]

.....

(iv) using ladybirds is less likely to damage the environment than using pesticides. [2]

.....

.....

.....



Examiner
only

- (b) (i) A farmer growing barley aims to harvest 8.0 tonnes/hectare.
Aphids reduce her harvest by 15 %.

Calculate the loss due to the aphids in tonnes/hectare. [2]

loss = tonnes/hectare

- (ii) The farmer is paid £117.00 per tonne for her barley.

Use your answer to (i) to calculate how much money the farmer loses per hectare
due to aphid damage. [1]

loss = £..... per hectare

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